ACTIVITY 3.3: DESIGN THE USER INTERFACES AND FINALISE THE SYSTEM DESIGN

What is a wireframe?

A wireframe is a simplified, visual representation of a screen or page that shows the layout, structure, and functionality of an interface without focusing on detailed design elements like colours or images.

Why should a wireframe be created before development?

It allows the team to plan and agree on the layout and functionality before spending time on coding. It saves time and money by catching problems early.

How can buttons, text fields and other controls be represented?

Buttons are shown as rectangles with labels; text fields as rectangles with placeholder text or lines; dropdowns as rectangles with arrows; checkboxes as small squares; and icons as simple shapes or placeholders.

Why should the first design focus on layout and functionality rather than colours?

Because the primary goal is to ensure the structure works and the user can complete their tasks. Colours and branding come later during the high-fidelity design stage.

How can a wireframe help identify design problems early?

By visualising the flow and layout, the team can spot missing elements, confusing navigation, or overcrowded screens before development begins, saving time and rework.

STEP 2: IDENTIFY THE REQUIRED FLUTTER SCREENS

FLUTTER SCREEN	USER	PURPOSE	RELATED REQUIREMENT / USE CASE
SPLASH / LOGIN	Parent/Guardian	Authenticate the parent and provide access to the app	NFR1, FR3 (Authentication)
CREATE ACCOUNT	Parent/Guardian	Allow new parents to register on the app	FR12 (User registration)
HOME (DASHBOARD)	Parent/Guardian	Provide quick access to main functions and latest news	FR1, FR2, FR3, FR5, FR6
MY CHILDREN	Parent/Guardian	View and manage children linked to the parent account	FR11 (Record-keeping)
CREATE CHILD PROFILE	Parent/Guardian	Add a child to the parent's account	FR11 (Record-keeping)
ATTENDANCE RECORDS	Parent/Guardian	View attendance summary and recent absences	FR2 (Attendance records)
REMINDERS	Parent/Guardian	Receive reminders for meetings and events	FR5 (Reminders)
MESSAGES	Parent/Guardian	View messages and announcements from teachers and admin	FR3 (Announcements)
CALENDAR	Parent/Guardian	View the school year plan and upcoming events	FR6 (Year plan)
DOCUMENTS	Parent/Guardian	Download report cards, letters, and forms	FR4 (Online statements)
PROFILE	Parent/Guardian	View and update personal contact information	FR12 (Update contact info)
EDIT	Parent/Guardian	Edit contact details	FR12 (Update

INFORMATION		and emergency contact	contact info)
DEMERIT TRACKING	Parent/Guardian	View learner demerit points and conduct history	FR9 (Points tracking)
NOTIFICATIONS	Parent/Guardian	View all app notifications in one place	FR5 (Reminders)
SETTINGS	Parent/Guardian	Manage app preferences and notifications	NFR6, NFR7 (Usability)

STEP 3: IDENTIFY THE REQUIRED ASP.NET SCREENS

PAGE	USER	PURPOSE	RELATED REQUIREMENT / USE CASE
LOGIN	Teacher / Admin	Authenticate staff members	NFR2 (Authorized staff)
DASHBOARD	Teacher / Admin	View summary stats and recent activity	FR8 (Manage communication)
CREATE ANNOUNCEMENT	Teacher / Admin	Create and send announcements to parents	FR3, FR7 (Announcements)
TRACK ENGAGEMENT	Teacher / Admin	View which parents have viewed announcements	FR7 (Track engagement)
RECORD POINTS	Teacher / Admin	Record merit points for learners	FR9 (Points tracking)
MANAGE USERS	Admin	Create, update, and delete user accounts	FR13 (Manage users)
GENERATE REPORTS	Admin	Generate communication and engagement reports	FR8 (Generate reports)
COMMUNICATION HISTORY	Teacher / Admin	View all past announcements and messages	FR10 (Store records)

STEP 5: SHOW NAVIGATION BETWEEN SCREENS

STARTING SCREEN	USER ACTION	NEXT SCREEN
SPLASH	Tap "Login"	Login
SPLASH	Tap "Create Account"	Create Account
LOGIN	Enter credentials → Tap "Log In"	Home (Dashboard)
LOGIN	Tap "Create Account"	Create Account
LOGIN	Tap "Forgot Password?"	Password Reset (future)
CREATE ACCOUNT	Fill form → Tap "Create Account"	Home (Dashboard)
CREATE ACCOUNT	Tap "Log In"	Login
HOME	Tap "Attendance Records"	Attendance Records
HOME	Tap "Reminders"	Reminders
HOME	Tap "Messages"	Messages
HOME	Tap "Merits"	Demerit Tracking
HOME	Tap Bottom Nav "News"	Messages
HOME	Tap Bottom Nav "Alerts"	Reminders
HOME	Tap Bottom Nav "Profile"	Profile
MY CHILDREN	Tap on child card	Child Profile / Academic Progress
MY CHILDREN	Tap "Add a Child"	Create Child Profile
ACADEMIC PROGRESS	Tap "Download Report"	(File Downloads)
ACADEMIC PROGRESS	Tap Back	Home
ATTENDANCE RECORDS	Tap "View Full Calendar"	Calendar
REMINDERS	Tap Back	Home
MESSAGES	Tap on message	Message Detail (future)
DOCUMENTS	Tap download icon	(File Downloads)
CALENDAR	Tap Back	Home
PROFILE	Tap "Edit Contact Information"	Edit Information
PROFILE	Tap "My Children"	My Children
PROFILE	Tap "Log Out"	Splash / Login
EDIT INFORMATION	Tap "Save"	Profile (updated)
EDIT INFORMATION	Tap Back	Profile

STEP 6: CHECK THE MOBILE AND WEB INTEGRATION

FLUTTER ACTION	SUPABASE DATA	ASP.NET ACTION
PARENT SUBMITS CHILD PROFILE	Child record created in children table	Teacher views learner record
PARENT VIEWS ATTENDANCE	Attendance records retrieved from attendance table	Teacher marks attendance
PARENT VIEWS ANNOUNCEMENTS	Announcements retrieved from announcements table	Teacher/Admin creates announcement
PARENT VIEWS REMINDERS	Reminders retrieved from reminders table	Teacher/Admin creates reminder
PARENT UPDATES PROFILE	User record updated in users table	Admin views updated parent details
PARENT VIEWS MERIT POINTS	Demerit records retrieved from demerit_records table	Teacher records new demerit points
PARENT VIEWS YEAR PLAN	Year plan retrieved from year_plan table	Admin updates year plan
PARENT VIEWS DOCUMENTS	Documents retrieved from documents table	Teacher/Admin uploads documents

STEP 7: CHECK YOUR DESIGN AGAINST THE ERD

INTERFACE INFORMATION	ENTITY /	ATTRIBUTE / FIELD

	TABLE	
PARENT NAME	User	FirstName, LastName
PARENT EMAIL	User	Email
PARENT PHONE	User	PhoneNumber
PARENT ADDRESS	User	Address
CHILD NAME	Student	FirstName, LastName
CHILD GRADE	Student	Grade
CHILD ID NUMBER	Student	IDNumber
CHILD PHOTO	Student	ProfilePicture
ATTENDANCE DAYS PRESENT	Attendance	DaysPresent
ATTENDANCE DAYS ABSENT	Attendance	DaysAbsent
ABSENCE REASON	Attendance	Reason, Date
ANNOUNCEMENT TITLE	Announcement	Title
ANNOUNCEMENT MESSAGE	Announcement	Message
ANNOUNCEMENT DATE	Announcement	DateCreated
REMINDER EVENT	Reminder	EventName, EventDate, EventTime, Location
MERIT POINTS	DemeritRecord	Points, Reason, DateIssued
DOCUMENT NAME	Document	Title, FileURL, DateUploaded
YEAR PLAN EVENT	YearPlan	EventName, StartDate, EndDate

STEP 8: CHECK ALIGNMENT ACROSS THE COMPLETE SYSTEM DESIGN

DESIGN ARTEFACT	QUESTION TO CHECK	EVIDENCE IN SKOOL-COMM
SRS	Is the function an approved requirement?	Yes — all FRs and NFRs are documented in the SRS.
ARCHITECTURE DIAGRAM	Does it show where the function operates?	Yes — Flutter (mobile), ASP.NET (web), Supabase (backend) are all mapped.
USE CASE DIAGRAM	Is the function available to the correct actor?	Yes — Parent, Teacher, and Admin each have specific use cases.
SEQUENCE DIAGRAM	Does it show how the important process occurs?	Yes — Flutter → Supabase → ASP.NET flows are documented.
ERD	Can the required information be stored?	Yes — All entities and attributes are defined in the ERD.
INTERFACE DESIGN	Can the user perform the function?	Yes — All screens are designed and connected via navigation.

Example Alignment : "Teacher must be able to record attendance"

ARTEFACT	EVIDENCE
SRS	Functional requirement FR15 exists — "Teachers must be able to record attendance after logging in."
USE CASE DIAGRAM	"Record Attendance" use case for Teacher, which is a child of "View Learner Progress" (generalisation)
SEQUENCE DIAGRAM	Teacher → ASP.NET → Supabase Database (same pattern as Process 2)
ERD	ATTENDANCE entity with attributes: AttendanceID, LearnerID, Date, Status, RecordedBy, DaysPresent, DaysAbsent, TotalDays
FLUTTER WIREFRAME	Parent sees the updated attendance summary on the Attendance Summary screen
ASP.NET WIREFRAME	Teacher marks learners present or absent on the Record Attendance page

STEP 9: OBTAIN STAKEHOLDER FEEDBACK

ITEM	DETAILS
DATE OF FEEDBACK	Friday ,09 October 2026
DESIGNS SHOWN	Splash, Login, Create Account, Home, My Children, Academic Progress, Attendance, Reminders, Messages, Calendar, Documents, Profile, Edit Information
STAKEHOLDER COMMENTS	“I am happy, I love the app and I think the school would appreciate this app”
CHANGES REQUESTED	Requested to remove academic progress, Attendance weekly record summary, Use the school Merit Template and to update the demerit point system
CHANGES MADE	Removed the academic and used the school Merit point template
EVIDENCE AVAILABLE	Meeting notes, Images, WhatsApp follow-up confirmation

Step 10: Update the System Design

DESIGN ARTEFACT	CHANGE MADE	REASON
ARCHITECTURE	Removed academic results and PDF statements; added attendance weekly summary,	Stakeholder changes
USE CASE DIAGRAM	Removed "Record Academic Results" and "Upload Academic Results"	Stakeholder changes
SEQUENCE DIAGRAM	Removed academic-results sequence	Alignment with new use cases
ERD	Removed ACADEMIC_RESULT; renamed BEHAVIOURAL_POINT to MERIT_POINT; added attendance summary fields, kept single STAFF table	Stakeholder changes and alignment
FLUTTER INTERFACE	Removed Academic; added Attendance Summary, Merit Points (school template)	Stakeholder changes
ASP.NET INTERFACE	Removed academic results upload; Attendance Summary	Stakeholder changes

STEP 11: UPDATE YOUR PROJECT RECORDS

RECORD	UPDATE NEEDED
MICROSOFT PROJECT	Mark design tasks as complete. Add development tasks for Phase 3.
GITHUB	Upload the latest SRS, ERD, wireframes, and architecture diagram.
PROJECT PROGRESS TRACKER	Update individual contributions (e.g., Lesego — wireframes + navigation flow).

STAKEHOLDER ENGAGEMENT	Add the feedback session date, attendees, and outcomes.
CHALLENGES AND ACTIONS	Document any design challenges and how they were resolved.